

A milliliter of salt

Senior Design Team Contract

University of Cincinnati

College of Education, Criminal Justice and Human Services

School of Information Technology

Team Members

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Intent

The following contract was written and agreed upon by Donald Durgan III, Christophe Vu, Connor York, Karl Isele. The contract provides expectations, objectives, and results for developing the game, "A milliliter of salt".

The contract is effective for all team members participating in the Senior Design Capstone class series in the 2026-2027 academic year.

Senior Design Contract

Project Summary

"A milliliter of salt" is a cooking game that explores the idiom of "You don't know what you have until it's gone." Often people take for granted, things that were mundane and abundant, not realizing how much they would miss it. The protagonist is exposed to the concept of that idiom through the loss of their mother. In order to move on, the protagonist takes up learning how to cook some of the meals that they used to share with the help of a friendly kitchen critter.

Problem Statement

Grief is a universal experience for everyone. The mundane tasks we share with loved ones, like making meals and baking sweets, often become the most missed memory after someone is gone. Zahara Gomez Lucini and the collective Las Rastreadoras del Fuerte, collaborated on a project to create a book containing articles, and photos about the missing peoples and enforced disappearance crisis in Mexico. This book also contains 27 recipes from the mothers of those victims. Stengel writes, "In the case of these mothers, their strength emanates from their love and, by conveying it through the recipe book, they tenderly touch the hearts of the readers" (Stengel 2025). Food can be an expression of love and compassion.

Being able to process feelings around grief is also important. Recently through interactive media like gaming, people can use this as a medium to confront loss. Eum and Doh writes that, "Our findings indicate that bereaved players recalled and related their autobiographical memories to their in-game experiences. Furthermore, they derived new meanings on both the loss and their post-loss life after playing video games. (Eum et al.) Through video games, players can connect their own situation and share that sense of loss through a game, recalling memories through various scenarios within the gameplay and story.

PS Works cited

Eum, Karam, and Young Yim Doh. "A Thematic Analysis of Bereaved Adults' Meaning-Making Experience of Loss through Playing Video Games." *Frontiers in Psychology*, vol. 14, 31 July 2023, www.ncbi.nlm.nih.gov/pmc/articles/PMC10426378/, <https://doi.org/10.3389/fpsyg.2023.1154976>.

STENGEL, NATALIA. "Recetario Para La Memoria." *Bulletin of Hispanic Studies*, vol. 102, no. 9, 16 Oct. 2025, pp. 999–1018, <https://doi.org/10.3828/bhs.2025.58>. Accessed 23 Apr. 2026.

Solution

A Milliliter of Salt is a solo or co-op game that explores the protagonist's journey through their grief due to the loss of their mother. Having never fully learned the recipes and dishes from their mother, they attempt to recreate those dishes from memory with the help of an unexpected tiny kitchen guest. Levels consist of 3 parts, the shopping phase (buying / getting ingredients), prepping phase (preparing the ingredients), and the cooking phase. The main player will go through a series of mini games for each phase, and the co-op player will assist by bringing ingredients to the main player or just messing around with the kitchen environment. Through their improvement in cooking, the player begins to accept the death of their mother by reminiscing about the memories that accompany certain dishes.

AMS will use Unity as the main game engine for development due to its extensive community support and resources. For source control, we will be either using Unity Version Control, or GITHUB to help facilitate collaborative work on this game.

Contact Information

TEAM MEMBER	DEGREE + TRACK TRACK	EMAIL	PHONE NUMBER OR OTHER CONTACT INFO?
Christophe Vu	BSIT – Software Dev	Vuvq@ucmail.uc.edu	513-375-1489
Donald Durgan III	BSIT – Game Dev & Simulation	Durgandl@mail.uc.edu	513-545-5334 (Text only)
Connor York	BSIT – Networking/Systems Administration & Game Design and Simulation	yorkco@mail.uc.edu	513-817-8143
Karl Isele	BSIT – Networking / Sys Admin & Data Technologies	iselekp@mail.uc.edu	513-807-3729

Project Source

Our team consists of members with varying backgrounds and interests. Some of us have a more technical background in programming, others have an interest in writing, and some with experience in 3d modeling and texturing. All members expressed interest in creating a game in order to best utilize all of our skills. The main concept of “A Milliliter of Salt” was created by Chris Vu as one of the potential game options.

Project Objectives/Goals

Project Objectives can be separated into three categories, Planning and Ideation, Development, and Publishing.

- *Planning and Ideation*
 - *Main story concept*
 - *UI/UX Design*
 - *Levels/Stage plans*
 - *Branding Guideline*
 - *Game Mechanics*
 - *Main Game Engine*

- *Project Roles*
 - *Technology Stack usage*
- *Development*
 - *Main Menu development*
 - *Game asset creation*
 - *Level design / implementation*
 - *Music / Sound design*
- *Publishing*
 - *Quality Assurance / Bug testing*
 - *Platform / build*

Main features of 'A ML of Salt'

- *3 Stage Levels*
 - *Procure: Grocery shopping*
 - *Prep: Preparing the ingredients*
 - *Cooking: Cooking phase*
- *Co-op option*
 - *2nd player plays as a kitchen critter to assist the main player*
- *Save/Load progress: With a multi-level game, players will need to be able to save their progress and resume at a later time.*
- *Story line / writing: Development of the underlying story is necessary to convey the full concept of the game. This will also help develop any cutscene / in game lines.*
- *Different game mechanics for 'cooking': To keep levels interesting, each level will have a varying type of gameplay to keep players interested.*
- *Scoring system for how well you 'cook' in each level: A scoring system of 0 – 100% for each level.*

Team Members and Responsibilities

Chris Vu – 3D Artist, Texture Artist, Programmer, Storyboard

Donald Durgan – Programmer, UI Design

Connor York – Programmer, Sound Design

Karl Isele – Programmer, Networking

Project Scope

Our team will develop a functional game application that enables users to experience cooking as a form of storytelling while exploring themes of memory and grief. The game, "A Milliliter of Salt," will guide players through a series of cooking levels that increase in complexity while also reflecting emotional progression tied to the story.

The scope includes designing and implementing 3-4 playable levels, each with unique cooking mechanics such as frying, mixing, timing, and ingredient selection. The team will develop core systems including player interaction, scoring (0–100%), level progression, and save/load

functionality. Additional features include a main menu, basic UI, and simple cutscenes or dialogue to support the storyline.

To meet the constraints of the Senior Design timeline, the team will focus on delivering a stable and complete core gameplay experience. Advanced features such as highly detailed animations, or large-scale environments will not be included. The goal is to produce a polished prototype that clearly demonstrates the concept, mechanics, and emotional narrative of the game.

Quick Project Timeline

<i>Task #</i>	<i>Task Name</i>	<i>Duration</i>	<i>Start Date</i>	<i>End Date</i>
1	Design Contract / Problem Research	1 week	August 31	September 6
2	Main Story Ideation	1 weeks	September 7	September 11
3	Technology Research/Procurement/Set-up	1 week	September 14	September 18
4	Level Ideation / Game Mechanic planning	3 weeks	September 21	October 9
3	Asset Creation / Modeling	5 Weeks	October 12	November 13
4	UI Development / Flow diagram	3 Weeks	September 21	October 9
5	Scene set up for Title/Menu, Level selection, Game scene	1 Weeks	October 9	October 16th
6	Menu Development	1 Week	October 9	October 16th
7	Level Selection Development	1 Week	October 9	October 16th
8	1 level "Procure/Grocery" mini game template	2 Week	October 16th	October 30th
9	1 level "Prep" mini game template	2 Week	October 16th	October 30th
10	1 level "Cooking" mini game template	2 Week	October 26th	November 6th
11	Presentation / DEMO	3 Week	November 16th	December 6th
12	2nd level "Procure/Grocery" mini game template	2 Week	December 7th	December 14th
13	2nd level "Prep" mini game template	1 Week	December 14th	December 21st
14	2nd level "Cooking" mini game template	1 Week	January 11th	January 18th

15	3rd level “Procure/Grocery” mini game template	1 Week	<i>January 18th</i>	<i>January 25th</i>
16	3rd level“Prep” mini game template	1 Week	<i>February 1st</i>	<i>February 8th</i>
17	3rd level“Cooking” mini game template	1 Week	<i>February 8th</i>	<i>February 15th</i>
18	“Stacked Dishes” mini game	1 week	<i>February 15th</i>	<i>February 22nd</i>
19	Audio Design and Background Music	<i>Persistent</i>	<i>October 9th</i>	<i>April 16th 2027</i>
20	Playtesting and Bug Fixing	<i>1 week</i>	<i>March 22nd</i>	<i>April 5th</i>
21	Adjustment / Feedback	<i>1 week</i>	<i>April 5th</i>	<i>April 13th</i>
21	Publishing	<i>1 day</i>	<i>April 16 2027</i>	

Technologies Used

The team will use Unity as the main game engine to develop the project because it supports 3D environments and scripting. Programming will be done using C# for implementing game mechanics and logic. Blender will be used for creating 3D models and textures for the game assets. Audio tools such as Audacity and REAPER will be used for sound effects and music editing. GitHub or Unity Version Control will be used for version control to allow team collaboration and track progress throughout development. Blender will be the main program used to create custom 3D Models for the setting/characters, and Substance Painter by Adobe will be used to create textures for those models. Lastly, the game will be hosted on Steam to enable online multiplayer (if players aren’t locally playing together).

Ethical Considerations

One ethical concern is the use of copyrighted materials. Copyright infringement in game development occurs when someone uses, reproduces, or distributes a developer's protected work without permission. This includes a game's code, artwork, music, storylines, and characters (ScoreDetect, 2024). Importantly, while gameplay mechanics and rules are generally not protected under copyright law, the audiovisual elements of a game, such as textures, models, and audio, are fully protected and cannot be reused without proper licensing (ScoreDetect, 2024). For this reason, all assets used in the project must either be created by the team or properly licensed to avoid legal issues and plagiarism. This ensures the project follows both university guidelines and industry standards, as game developers have the right to enforce their copyrights through demands for removal of infringing content, damages, or injunctions (ScoreDetect, 2024).

Another consideration is how the game represents the theme of grief. Research combining the expertise of games studies scholars, medical ethicists, and clinical psychologists notes that grief is a universal human experience, but navigating its psychological, social, and embodied effects can be a challenging task (Stones in Our Pockets, 2024). Since video games communicate across multiple modes, they carry significant responsibility in how they portray loss. The paper cautions that "aiming for ultra-realism in depicting loss could potentially re-

traumatise players" and instead recommends using narrative and visual elements that empower players to construct their own meanings (Stones in Our Pockets, 2024). Because the story is based on losing a loved one, the team must present this topic in a respectful and thoughtful way. The goal is to create an emotional experience without being offensive or insensitive to players who may personally relate to the topic.

The team must also ensure fair collaboration. Each member is responsible for completing their assigned tasks and communicating with any issues. Ethical teamwork helps maintain trust and ensures the project is completed successfully.

Ethical Works Cited

Imri, & Sumner, M. (n.d.). SCOREDETECT team. Copyright Compliance in the Gaming Industry: An Overview.
<https://www.scoredetect.com/blog/posts/copyright-compliance-in-the-gaming-industry-an-overview>

Reay, E., Ma, M., Mankee-Williams, A., Pavarini, G., Shaughnessy, N., & Bhui, K. (2024). Stones in Our Pockets: Mental Health Dimensions of Grief in Contemporary Video Games.
<https://doi.org/10.1101/2024.04.23.24306235>

Market Analysis Research

Direct Competitors

A cooking game isn't new or revolutionary within the industry, primary examples being Cooking Mama made for the Nintendo DS, or the flash game "Papa's Pizzeria". These games have more of a focus on game play mechanics rather than any deeper narrative. Two games that fall under the same our game's genre of 'narrative cooking' are Venba, an Indian cooking game developed and published by Abhi Swaminathan, and A Taste of the Past, a Chinese cooking game made by Sondering Studios. Venba follows an Indian family that immigrated to Canada, and through cooking cultural dishes from a recipe book. It explores the family's experiences like financial hardship, discrimination, and language barriers, issues that immigrant families often face. Like a Milliliter of Salt (AMS), Venba focuses on the emotional narrative centered around family, but AMS is more aimed towards processing the loss of family through cooking. A Taste of the Past has the greatest overlap with AMS, having a similar focus of handling grief and emotions of losing a loved one, and exploring that through un-learned recipes. The main difference between our games is the artistic direction. A Taste of the Past is a 2D Side scrolling game, whereas AMS will utilize 3D models for the environment and player models.

Medical Treatment and Games

Games are becoming used in evidence-based professional intervention, especially in mental health, rather than only for amusement. The use of the video game Tetris as a "cognitive vaccine" against post-traumatic stress disorder (PTSD) is one notable example. A brief digital intervention involving guided Tetris gameplay significantly reduced intrusive memories of trauma, with 70% of participants reporting no flashbacks six months later, according to a randomized controlled experiment with frontline healthcare workers. The treatment demonstrates a targeted application of cognitive science in a scalable, low-intensity digital therapy by employing the visuospatial demands of the game (mental rotation of blocks) to compete with and limit the brain's capacity to store vivid, emotional trauma imagery.

Video games are being incorporated into more comprehensive treatment frameworks in addition to symptom-targeted tools like Tetris. For example, Video Game Therapy® (VGT®) approach centers psychological therapy on the gaming experience, use commercial games to promote emotional expression, develop coping mechanisms, and establish a secure setting for trauma exploration. With applications ranging from enhancing social skills in children with autism spectrum disorder (ASD) to serving as a medium for exposure therapy in phobias and PTSD, this approach uses the immersive "flow" state of gaming to assist patients in processing emotions and behaviors. The method is customized for each patient, often utilizing instruments such as the Myers-Briggs Type Indicator (MBTI) to choose games that best suit their personality.

Games are also used in the medical field to cure a variety of illnesses and aid in cognitive rehabilitation. According to research, playing video games can help people with ADHD, traumatic brain injuries, and neurodegenerative diseases with their focus, problem-solving, and social skills. For instance, the FDA has authorized the action video game EndeavorRx as a prescription digital treatment for ADHD in children. Additionally, commercial driving simulators

and games like "Full Spectrum Warrior" have been employed as "in virtuo" exposure treatment tools to assist patients in safely confronting and managing phobias and trauma-related triggers, such as those from military service or auto accidents.

The field strikes a mix between hope and caution despite its promise. Larger, more varied clinical trials are needed to further investigate the long-term efficacy and wide applicability of therapies such as Tetris-based therapy. The Video Game treatment® approach also recognizes its limitations, stating that it is a supplement to traditional treatment rather than its replacement and that it must be used carefully by a qualified therapist to prevent negative outcomes. The main objective is to provide user-friendly, affordable, and entertaining digital therapies that can reach people without access to traditional, resource-intensive mental health care.

Grief & Loss in Gaming

In games covering topics of grief, loss, and trauma, those experiences and emotions are conveyed to the player in the form of game mechanics, visual art and/or audio, and through the game's narrative. Games utilize gameplay mechanics to invest players in the experience and to grant them a connection to the themes of loss. The game *What Remains of Edith Finch* by developer Giant Sparrow utilizes gameplay to let players experience the life each family member on the day of their death as Edith explores the rooms of her family home for the first time in seven years. These gameplay sections let the player directly connect with these experiences as they learn the fate of each family member. By placing the player in control of these minigames, the game effectively invests the player in the outcome of that gameplay, as it all led to each member of the Finch family's passing.

Art within a game effectively surrounds a player in the emotions a developer wishes to portray. Representing grief through art in a game can take many forms, such as somber music, dark visuals, and so on. The game *GRIS* by developer Nomada Studios utilizes art as a representation of the process of emotional growth and moving beyond grief. The game starts in a world lacking color save for gray and black. As the player progresses, the world grows more colorful, and more abilities are unlocked, which is all used to represent the journey out of grief and the beauty that can be found beyond it.

A game's narrative sets the tone for the entire experience and guides the player along the journey. The narrative can either be direct or indirect in the way that it represents grief, in the case of *GRIS* it's indirect as there is no dialog or text directly setting up the narrative tone, and it's all guided by the environment. The game *That Dragon, Cancer* by developer Numinous Games takes the opposite extreme approach and uses heavy direct narrative to tell the story of Joel and his family, as they navigate his battle with cancer and ultimately his passing. The game takes place over a series of vignettes that capture moments of Joel's and his family's life. Each section has little gameplay and instead uses direct narration to tell an interactive story. The narrative of any game covering a topic like grief will represent the emotional journey that a developer wishes to put the player through.

Grief & Loss in other forms of Media / Entertainment

Joan Didion's *The Year of Magical Thinking* discusses how grief can push someone to delusions, such as denying the reality of losing someone. Didion also explains how life gets in

the way of processing the loss of someone, for example, dealing with cold social interactions regarding her husband's death, like paperwork, and caring for her hospitalized daughter. This reveals the harshness of life and how we cope with it. Our game can express these experiences via the need to feed yourself despite crippling grief and magical thinking such as "if I can recreate my mother's dish, then she'll come back to me".

Green Day's *Wake Me Up When September Ends* is a song about the death of Billie Joe Armstrong's father. Armstrong explains that the song is a way of honoring his father's memory during his process of mourning: "I think about him every day, really. I kinda avoided writing about him for many years, and then finally having a breakthrough like that felt good. It wasn't like a negative emotion so much, but it was just kind of like honoring him." This same process will be shown in *AMS* with the protagonist honoring their mother's memory through cooking.

Research Works Cited

Andersson, Erik, et al. "Innovations in Digital Interventions for Psychological Trauma: Harnessing Advances in Cognitive Science." *mHealth*, U.S. National Library of Medicine, 19 Oct. 2018, [pmc.ncbi.nlm.nih.gov/articles/PMC6232069/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC6232069/).

Bocci, Francesco, et al. "Putting the Gaming Experience at the Center of the Therapy-the Video Game Therapy® Approach." *Healthcare (Basel, Switzerland)*, U.S. National Library of Medicine, 15 June 2023, [pmc.ncbi.nlm.nih.gov/articles/PMC10298057/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC10298057/).

Gris, www.devolverdigital.com/games/gris. Accessed 19 Sept. 2026.

Hopper, Alex. "The Meaning behind 'Wake Me up When September Ends' by Green Day." *American Songwriter*, 7 June 2022, americansongwriter.com/the-meaning-behind-wake-me-up-when-september-ends-by-green-day/.

"Mental Health Treatment That Includes Tetris® Gameplay Dramatically Reduces Traumatic Flashbacks | News | Wellcome." *Wellcome*, 18 Feb. 2026, wellcome.org/insights/articles/mental-health-treatment-includes-tetrisr-gameplay-dramatically-reduces-traumatic.

Pinsky, Robert. "'The Year of Magical Thinking': Goodbye to All That (Published 2005)." *The New York Times*, The New York Times, 9 Oct. 2005, www.nytimes.com/2005/10/09/books/review/the-year-of-magical-thinking-goodbye-to-all-that.html.

"That Dragon, Cancer." *That Dragon, Cancer*, www.thatdragoncancer.com/. Accessed 18 Sept. 2026.

"What Remains of Edith Finch." *Annapurna Interactive*, annapurnainteractive.com/games/what-remains-of-edith-finch. Accessed 18 Sept. 2026.

Team Rules

1. Let Donald know 2 days, or before 4pm **immediately**, before the official meeting day so he can request interpreters for the meetings.
2. Team members, that have less classes or more free time, shouldn't do more than anything they are required to do or taking over others tasks without checking if they need help

3. *Plagiarism will not be tolerated. Any team member that plagiarizes will be subject to university policies and a team meeting will be called.*
4. *Each team member will stay current on their tasks to ensure the project milestones are being met. If an event conflicts that will affect the completion of a deliverable, the team member will notify the other team members at least 24 hours in advance of the scheduled due date.*
5. *If a group member will be absent on class days or for an extended period of time, they will notify both the other team members and the Instructors.*
6. *Advisor Meetings will be every other Monday after lecture with Samuel Bricking (first come first serve) Typically 30 minutes for the first semester.*

Team Signatures:

Signature : Donald Durgan III

Date: 4/1/2026

Signature : Vuong Christophe Vu

Date: 4/10/26

Signature : Connor York

Date: 4/12/26

Signature : Karl Isele

Date: 9/3/2026

Advisor Signature:

Date: 9/21/2026

Signed by:
Samuel Bricking
CB00DE8A3A3D48F...